

Image Classification of Covid-19

42048 STUDIO 3: INNOVATION

ASSESSMENT TASK 3: RESEARCH PAPER

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Table of Contents

1. Introduction	2
1.1 Introduction to the Business Context	2
1.2 Problem Statement	2
1.3 Motivation.....	3
1.4 Project Contributions.....	3
2. Scoping the Project.....	5
2.1 Project Aim.....	5
2.2 Objectives and Deliverables	5
2.2.1 Objectives.....	5
2.2.2 Deliverables:.....	6
3. Methodology	9
3.1 Methodology Framework.....	9
3.2 Data Understanding.....	9
3.3 Data Preparation	11
3.4 Exploratory Data Analysis.....	14
3.5 Model Selection.....	17
4. Experiments	20
4.1 Experiment Plan.....	20
4.2 Dataset Preparation.....	21
4.3 Model Development	22
4.4 Model Training	23
4.5 Summary of Learning Curves.....	24
4.6 Model Validation and Evaluation	25
4.7 Is there scope for Improving the model?	26
5. Discussions and Conclusion.....	27
5.1 Possible Limitations	27
5.2 Future recommendations.....	28
References	30
Appendix	31
Contribution Table	31

1. Introduction

1.1 Introduction to the Business Context

The global outbreak of COVID-19 has posed unprecedented challenges to healthcare systems worldwide, exposing the limitations of traditional diagnostic tools in terms of sensitivity, cost, and turnaround times. Current primary methods for COVID-19 diagnosis, such as Serology, RT-PCR, and Antigen testing, struggle with issues like limited sensitivity and long processing times (Simonite, 2020). In response, there is a pressing need for innovative solutions that can enhance diagnostic efficiency and accuracy, particularly in high-demand and resource-limited settings.

The research from Northwestern University's Feinberg School of Medicine revealed that secondary bacterial pneumonia was a significant factor in many COVID-19 deaths, especially among patients requiring mechanical ventilation. Nearly half of these patients developed ventilator-associated bacterial pneumonia. The study highlighted that unresolved secondary pneumonia was a major cause of mortality, more so than the viral infection itself. Furthermore, the findings challenged the common belief that a "cytokine storm" was a major cause of death in these patients (Paul, 2023).

The study further underscored that these secondary infections, rather than the primary viral infection, often dictated the clinical outcomes of patients, emphasizing the need for strategies focused on preventing and treating bacterial pneumonia alongside COVID-19. It also suggested that the treatment protocols at the time, which were heavily geared towards managing the viral aspects of COVID-19, may have overlooked the critical impact of bacterial complications, thereby pointing towards a potential shift in treatment strategies for future respiratory pandemics (Paul, 2023).

1.2 Problem Statement

Despite advancements in medical technology, the diagnosis of COVID-19 remains a critical bottleneck in managing the health crisis effectively. The prolonged turnaround times and the high costs associated with traditional diagnostic methods hinder rapid patient management and effective control of the virus spread (Simonite, 2020). Consequently, the development of an automated detection system using chest X-rays (CXR) and CT scans presents a promising avenue to address these challenges.

1.3 Motivation

This project is driven by the imperative to improve health outcomes and resource allocation during outbreaks. With COVID-19 and pneumonia still prevalent worldwide, enhancing diagnostic processes is not only crucial for current pandemic management but also for preparedness against future outbreaks (Aggarwal et al., 2022). The use of deep learning in medical imaging has shown great potential in other domains and applying this technology to detect COVID-19 could significantly reduce the burden on healthcare systems by providing faster and more accurate diagnoses.

1.4 Project Contributions

1. This research paper on machine learning follows a structured format, thoughtfully adapted from the widely recognized CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology to cater specifically to business audiences. The adaptation is illustrated in the table below, which aligns each section of the paper with the corresponding stages of the CRISP-DM process.
2. Introduction: This initial section sets the stage for the paper by covering the 'Scoping the Project' and 'Business Understanding' stages of CRISP-DM. Here, the primary focus is on defining the project's scope and understanding the business context and requirements.
3. Methodology: This part corresponds to the 'Data Understanding' and 'Data Preparation' stages. It delves into the methods used for collecting, understanding, and preparing the data for analysis, providing a foundation for the subsequent modelling.
4. Experiments: Aligned with the 'Modelling' stage of CRISP-DM, this section details the modelling approaches and techniques employed to analyse the data. It is critical in demonstrating how theoretical models are practically applied to derive insights.
5. Evaluation: This section corresponds directly to the 'Evaluation' stage, discussing how the models and their results are assessed against predefined criteria to ensure validity and reliability of the findings.
6. Discussion and Conclusions: The final part of the paper maps to the 'Deployment' stage of CRISP-DM. It encapsulates the discussions on the implications of the findings, how they can be applied in practice, and the conclusions drawn from the research.

3 main areas:

Innovative Application of Deep Learning Models: By employing Convolutional Neural Networks (CNNs), this project seeks to develop cutting-edge models that can differentiate COVID-19 from other respiratory conditions such as MERS, SARS, and ARDS based on CXR and CT images (Aggarwal et al., 2022).

Methodology: The research will utilize advanced image processing techniques and deep learning algorithms to train models that identify unique visual signatures of COVID-19 pneumonia, potentially setting a benchmark for accuracy in automated medical diagnosis tools (Aggarwal et al., 2022).

Experimental Results: The project will rigorously evaluate the performance of the developed models against existing diagnostic methods, aiming to demonstrate improvements in speed, cost-efficiency, and diagnostic accuracy.

2. Scoping the Project

The COVID-19 outbreak, had a major impact globally. Traditional diagnostic methods for COVID-19, such as RT-PCR, antigen testing, and serology, were helpful however had their drawbacks. These tests were expensive, took longer time, and may not always be completely accurate, because of this, there was an imperative need to develop automated detection systems for COVID-19. Therefore, this aim of this project is to leverage deep learning and image classification techniques to enhance the efficiency and accuracy of diagnosing COVID-19 through chest X-rays and CT scans.

2.1 Project Aim

The image classification system goes beyond a simple X-ray. It uses machine learning and deep learning to analyze the image and tell the difference between COVID-19, other lung infections, and healthy lungs. The aim is to create a dependable tool for medical professionals that can help them in diagnosing COVID-19 earlier and more accurately. By doing this, it can improve patient care, lighten the load on healthcare systems and support healthcare professionals in making more precise diagnoses.

2.2 Objectives and Deliverables

The main objective of this project is to develop a deep learning model that can classify the images for detecting COVID-19 and leveraging Convolutional Neural Networks (CNNs) that will train the deep learning algorithm to identify visual patterns in the given images.

However, in this report the entire process of developing the model has been classified into small objectives that has to be fulfilled to achieve the main aim of the project.

2.2.1 Objectives

- **Data Collection and Preparation**

The first step includes a vast collection of chest X-ray images. This collection will include images from people with COVID-19, other lung infections, and healthy lungs. To make this data more reliable, we'll clean it up, augment it, and ensure it's all on the same scale. This will prepare the data for the training and testing stage.

- **Model Development**

After data collection, a powerful deep learning model will be developed. This model will potentially look at the chest X-rays and CT scans and accurately tell if someone has COVID-19. It also differentiates COVID-19 from other lung problems like MERS, SARS, and ARDS. To achieve this, we will be leveraging a type of deep learning called Convolutional Neural Networks (CNNs). CNNs are great at recognizing patterns in images. The model will be trained to find the unique visual signatures of COVID-19 pneumonia in medical scans.

- **Validation and testing**

Next is to make sure how it works well in the real world. For this the model needs to be validated by using a separate testing dataset and then testing the same from different sources (external validation). This ensures that it can handle variations in real-world X-rays and CT scans.

- **Deployment**

A user-friendly interface will be developed for the classification system which will make it simple for the medical professionals to use. Later, the model will be integrated into a software application that can be used in hospitals and clinics.

- **Documentation and Reporting**

At the end, a detailed record of the entire process will be documented. This will include information like where we got the data, how the model is built, how we trained it, and how well it performed. A comprehensive report will be prepared to explain the findings and show how effective the system is. This will help others understand and potentially use the work.

2.2.2 Deliverables:

- A curated dataset of annotated chest X-ray images.
- A trained and optimized deep learning model for COVID-19 classification.
- A user-friendly application for deploying the model in a clinical setting.
- A detailed project report documenting the methodology, results, and conclusions.

2.3 Stakeholder Analysis

High-Interest, High-Influence Stakeholders

- **Chief Medical Officer (CMO):**

The CMO will ensure the project aligns with best practices and improves patient outcomes. They'll provide medical expertise and guidance, making sure the project fits seamlessly with the hospital's goals and standards.

Medium-Interest, Low-Influence

- **Radiologists and Physicians:**

These medical professionals are crucial for validating the data and the model's accuracy. Their feedback will ensure that the system is clinically relevant and effective for diagnosing COVID-19.

- **Department Heads (Radiology, IT):**

These stakeholders are key to integrating the model into daily workflows. They'll provide resources and support for data collection and ensure high-quality image analysis.

Low-Interest, Low-Influence

- **Information Technology (IT) Department:**

The IT team will be responsible for integrating the model into the hospital's existing systems, ensuring compatibility and security for smooth operation.

- **Quality Improvement and Patient Safety Teams:**

While not directly involved in development, these teams will be vital for monitoring the project's impact and ensuring ongoing improvement aligns with quality and safety standards. Their feedback is essential for success.

2.4 Defining Success Criteria

Accuracy and Reliability

The model's accuracy in correctly identifying COVID-19 from chest X-ray and CT images is the primary success criterion. High accuracy reduces false positives and negatives, ensuring reliable diagnostic support.

Operational Efficiency

The implementation of an automated screening tool should streamline the diagnostic process, reducing the workload on medical staff and expediting patient care. Efficiency is measured by the reduction in diagnostic time and the increased capacity for handling patient volume.

Cost Savings and Resource Optimization

The model should contribute to significant cost savings by reducing the need for multiple traditional tests. Resource optimization involves effective allocation of medical resources, ensuring critical cases receive timely attention.

Improved Patient Outcomes

Ultimately, the success of the project is gauged by improved patient outcomes, including faster diagnosis, timely treatment, and reduced mortality rates. Enhanced diagnostic accuracy leads to better clinical decisions and patient management.

Therefore, by developing an accurate and efficient automated screening tool, the project seeks to enhance diagnostic capabilities, optimize healthcare resources, and ultimately improve patient outcomes. Stakeholder engagement and clear success criteria will guide the project's execution, ensuring alignment with clinical standards and hospital objectives. Through careful planning, data analysis, and model development, this initiative holds the potential to significantly impact the fight against COVID-19.

3. Methodology

3.1 Methodology Framework

Our project employs the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, a robust framework for developing data mining projects. This framework includes six stages: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment (Hotz, 2024). For this project, we also incorporated Agile methodologies to ensure increased collaboration, rapid stakeholder feedback, and iterative improvements. A brief breakdown of the steps can be seen below in Figure 1.

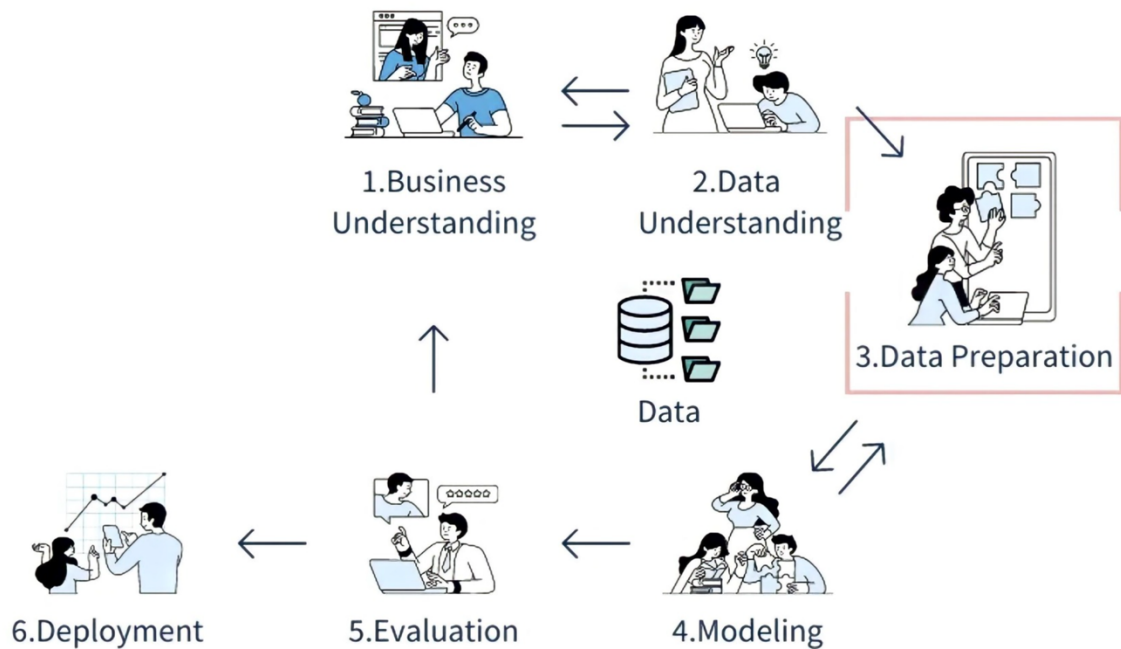


Figure 1: CRISP-DM

The Covid-19 chest Xray data was obtained from Kaggle and is the main source for building the machine learning model. The following sections will delve into a detailed analysis of the data and describe the preparation steps undertaken to enhance its suitability for model implementation.

3.2 Data Understanding

3.2.1 Describing Data

The COVID-19 chest X-ray dataset obtained from Kaggle is the primary source for building our machine learning model. This dataset includes a collection of chest X-ray images along

with 29 associated metadata that provides additional context for the images, such as patient ID, age, sex, and type of pneumonia. The primary objective is to accurately classify these images to detect COVID-19 (Bachir, 2020).

There are 372 images in total, but the metadata has varying numbers of rows due to many missing values across different metadata fields. For instance, some fields like offset, sex, age, survival, date, and location have significant missing values, which were handled during data cleaning.

The metadata includes various types of variables:

- Numerical Variables: Examples include patient age and the number of days since symptom onset (offset).
- Categorical Variables: Examples include patient sex, type of pneumonia (finding), and the type of X-ray view (view).
- Text Variables: Examples include patient ID, filename, and location information.

RangeIndex: 372 entries, 0 to 371
Data columns (total 29 columns):

#	Column	Non-Null Count	Dtype
0	patientid	372 non-null	int64
1	offset	276 non-null	float64
2	sex	329 non-null	object
3	age	318 non-null	float64
4	finding	372 non-null	object
5	survival	116 non-null	object
6	intubated	72 non-null	object
7	intubation_present	77 non-null	object
8	went_icu	35 non-null	object
9	in_icu	7 non-null	object
10	needed_supplemental_o2	12 non-null	object
11	extubated	23 non-null	object
12	temperature	35 non-null	float64
13	p02_saturation	44 non-null	float64
14	leukocyte_count	11 non-null	float64
15	neutrophil_count	2 non-null	float64
16	lymphocyte_count	10 non-null	float64
17	view	372 non-null	object
18	modality	372 non-null	object
19	date	291 non-null	object
20	location	254 non-null	object
21	folder	372 non-null	object
22	filename	372 non-null	object
23	doi	104 non-null	object
24	url	372 non-null	object
25	license	206 non-null	object
26	clinical_notes	362 non-null	object
27	other_notes	234 non-null	object
28	Unnamed: 28	4 non-null	object

Figure 2. All metadata variables

Out of these 29 variables, 11 were identified as particularly useful. For the primary purpose of identifying X-ray images of COVID-19 patients, the key variables are the picture file and

the findings. Additionally, other variables provide valuable insights into the demographics of the patients.

#	Column	Non-Null Count	Dtype
0	patientid	372 non-null	int64
1	offset	276 non-null	float64
2	sex	329 non-null	object
3	age	318 non-null	float64
4	finding	372 non-null	object
5	survival	116 non-null	object
6	view	372 non-null	object
7	modality	372 non-null	object
8	date	291 non-null	object
9	location	254 non-null	object
10	filename	372 non-null	object

Figure 3. Key metadata variables used for analysis

Below is a brief description of these 11 key variables.

FEATURES	DESCRIPTION
patientid	Internal identifier to identify which image belongs to which patient
offset	Number of days since the start of symptoms or hospitalization for each image. If a report indicates "after a few days", then 5 days is assumed. This is very important to have when there are multiple images for the same patient to track progression.
sex	Male (M), Female (F), or blank
age	Age of the patient in years
finding	Type of pneumonia
survival	Yes (Y) or No (N) or blank if unknown
view	Type of image; for example, PA, AP, or L for X-rays and Axial or Coronal for CT scans
modality	CT, X-ray, or something else
date	Date the image was acquired
location	Hospital name, city, state, country
filename	Name of the image file

Table 1. Key metadata variables

3.3 Data Preparation

3.3.1 Data Cleaning

Before starting the cleaning process, we start by getting a view of the first row of our data. It can be examined that the location variable includes the whole address whereas we only need the country for our analysis. As the whole address is not beneficial, we will extract the country from each address and replace every location with just the country name.

```

patientid      2
offset         0.0
sex            M
age            65.0
finding        COVID-19
survival       Y
view           PA
modality       X-ray
date           January 22, 2020
location       Cho Ray Hospital, Ho Chi Minh City, Vietnam
filename       auntminnie-a-2020_01_28_23_51_6665_2020_01_28_...

```

Figure 4. Values of the first row of the dataset

After successful implementation of the code the desired results were achieved

Result:

location	Vietnam
----------	---------

The next step is to fix datatypes. From figure 3, we can see the datatypes of our variables. It can be observed that date is treated as an object datatype. Therefore, it would be better to give it the 'date' datatype. The benefits of this includes easier coding when building visualizations and a better understanding of the data.

Result:

```

8  date      288 non-null  datetime64[ns]  ----> Datatype

date                2020-01-22 00:00:00  ----> Table output

```

Furthermore, from Figure 3, we can also observe the numerous missing values through the 'Non-Null Count' column. Figure 5 visualizes these missing values by showing which columns have what percentage of missing values.

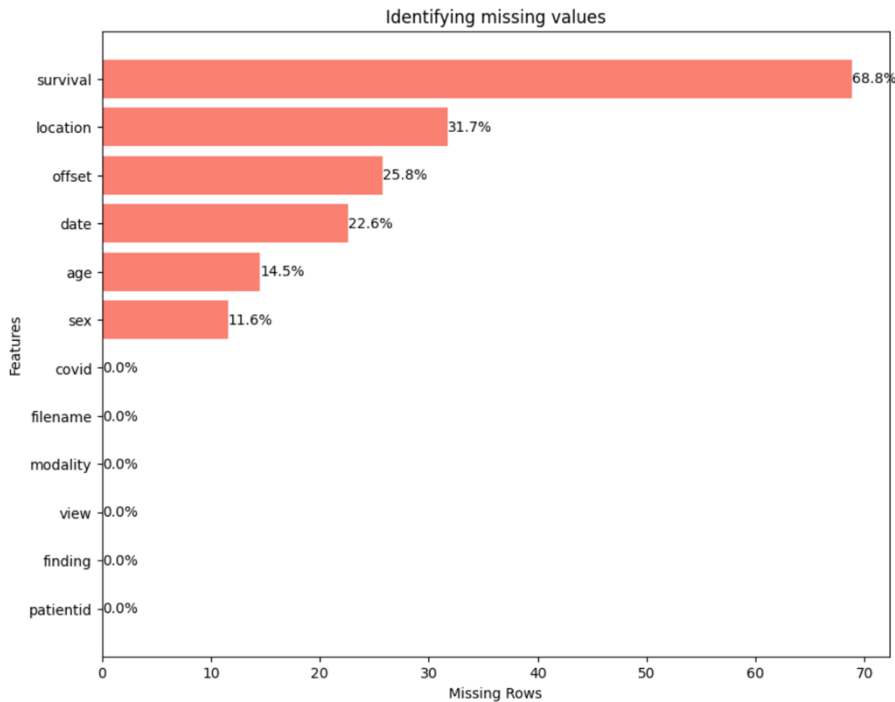


Figure 5. Percentage of Missing values

Due the huge number of these missing rows, it can be difficult to resolve them by taking averages. Furthermore, as we wish to use as many pictures as possible for our machine learning model, removing any number of rows is not a good idea. Therefore, as our independent variable mainly constitutes Xray pictures, it can be safe to remove all columns with missing values. However, in the data exploration stage these variables can still be used to give a general idea of our demographics.

3.3.2 Data Transformation

For data transformation, there was only one variable created, however this variable holds a lot of value as it is the dependent variable. The Covid variable is a binary variable, one-hot encoded as 1 or 0 with 1 indicating that the ‘finding’ of the Xray was covid positive and 0 indicating that it was not.

After the successful implementation of the code, the covid variable was added to the table. Counting the number of covid positive and negative cases, the below results were observed.

Result:

count	
covid	
1	308
0	64

3.4 Exploratory Data Analysis

Exploratory data analysis is done to understand the dataset better. This can include understanding the variables and their distributions. Furthermore, a preliminary analysis can be done to identify relationships between our independent and dependent variables.

As the dataset only includes pictures as the independent variable, a preliminary analysis is not viable. Therefore, the metadata is used to understand the demographics of the people that provided the Xray images.

1. Finding the distribution of offset days

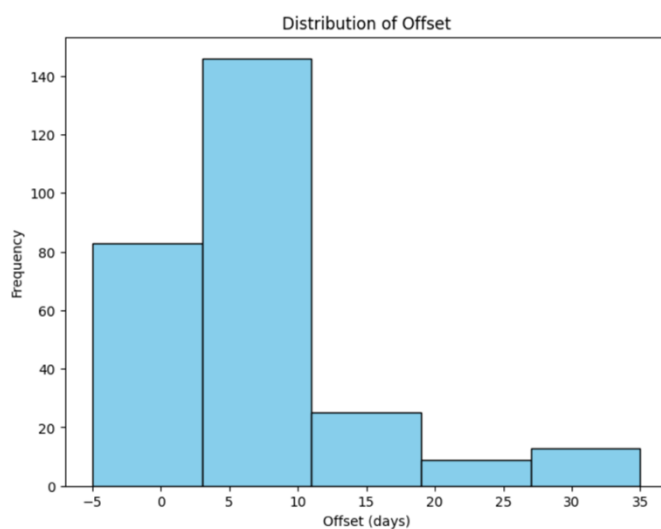


Figure 6. Distribution of offset days

Visualization: Histogram

Observation: The histogram shows that most cases had an offset period of 5 to 10 days. This shows that they were hospitalized or started showing symptoms 5 to 10 days before their Xray.

2. Finding the distribution of sex

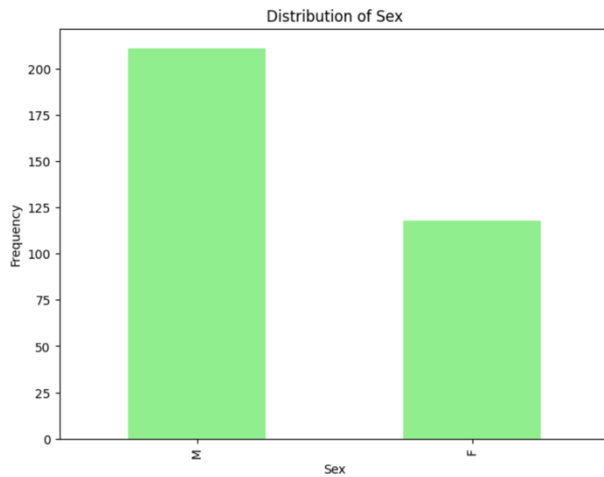


Figure 7. Distribution of Sex

Visualization: Bar chart

Observation: The visualization indicates that most of the Xray images come from males.

3. Finding the survival rate

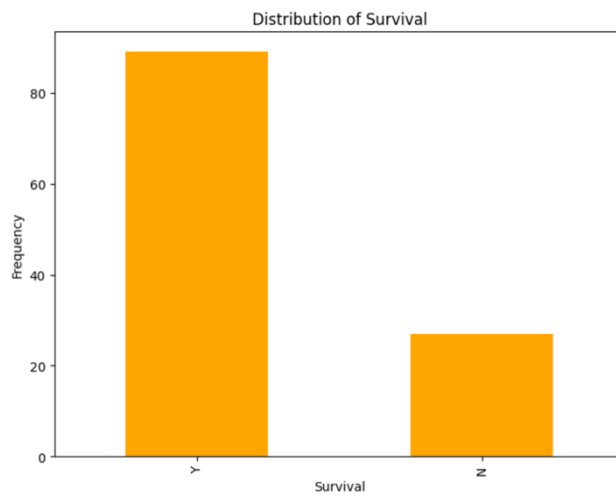


Figure 8. Distribution of survival rate of the patients

Visualization: Bar chart

Observation: Most of the patients survived.

4. Where do the Xray images come from

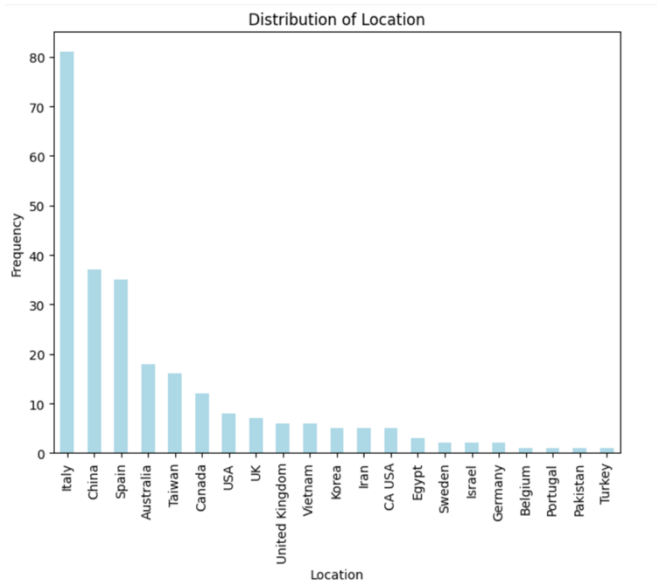


Figure 9. Location of the data source

Visualization: Bar chart

Observation: From our location chart we can ascertain that most X-ray images come from Italy, with China coming in second and Spain a close third.

5. Overtime images acquired

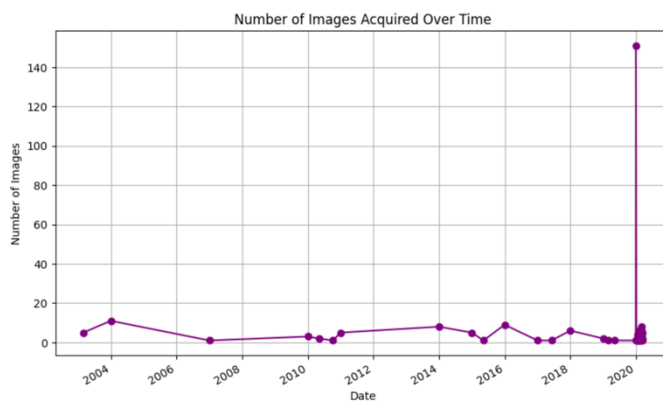


Figure 10. Number of images acquired over time

Visualization: Line chart

Observation: This chart shows that the images have been collected since around 2003 however most of them come from 2020 as observed from the spike.

6. Finding out how age is distributed

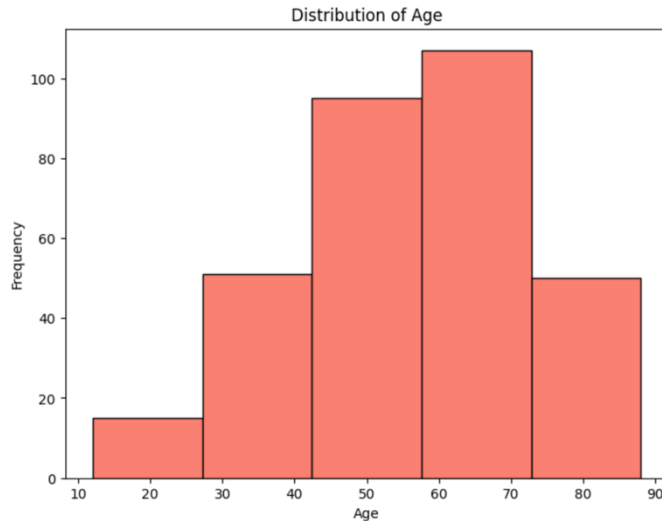


Figure 11. Age distribution of the patients

Visualization: Histogram

Observation: This chart shows that majority of the xray scans come from people aged between 40 and 70

3.5 Model Selection

The goal of our classification model is to accurately detect COVID-19 from chest X-ray images. Various machine learning algorithms can address this image classification problem, including Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), Decision Trees, and k-Nearest Neighbors (k-NN). Each classifier is suitable for different contexts based on data characteristics and the specific requirements of the task. For our project, CNNs were chosen due to their superior performance in handling image data and capturing spatial hierarchies.

Convolutional Neural Networks (CNNs)

CNNs are a type of deep learning model specifically designed for image classification tasks. They are capable of automatically and adaptively learning spatial hierarchies of features through backpropagation. CNNs consist of several types of layers, including convolutional layers, pooling layers, and fully connected layers (Figure 14).

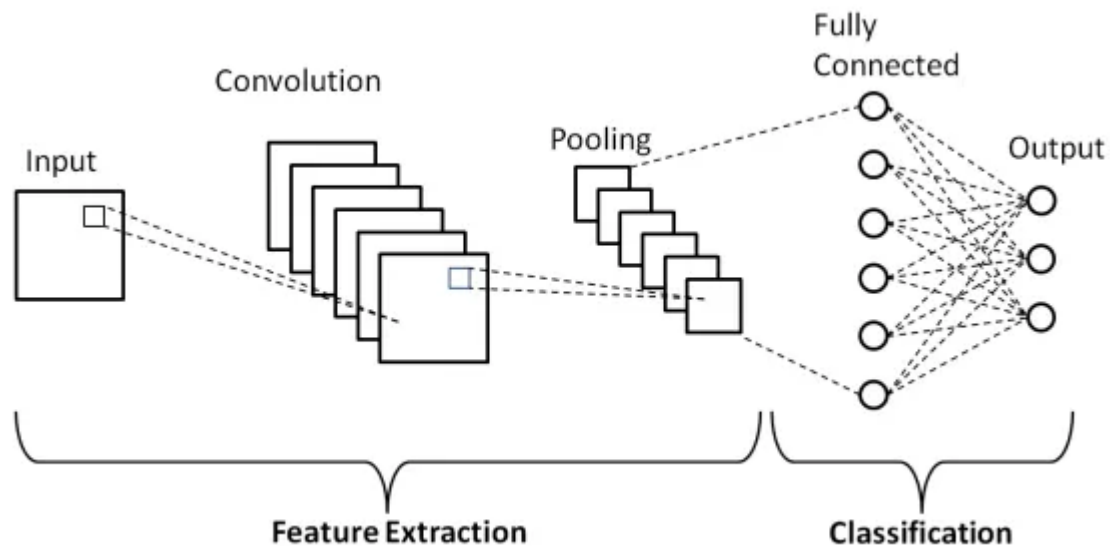


Figure 12. Architecture of a CNN (Fan & Truong, 2022)

Rationale for Choosing CNNs:

- **Feature Learning:** CNNs are designed to automatically learn and extract features from images, which is crucial for identifying patterns associated with COVID-19 in chest X-rays. This reduces the need for manual feature engineering.
- **Handling Spatial Hierarchies:** CNNs excel at capturing spatial hierarchies in images through convolutional layers that detect edges, textures, and more complex patterns. This capability is essential for analysing medical images where subtle details can be important for diagnosis.
- **High Accuracy:** CNNs have demonstrated high accuracy in various image classification tasks. For medical image analysis, achieving high accuracy is critical, and CNNs have a proven track record in this domain.

Comparison with other models

- **Support Vector Machines (SVMs)**

SVMs can be adapted for image data using specialized kernels like Radial Basis Function (RBF).

SVMs are effective with smaller datasets and robust against overfitting with appropriate kernels. However, they require manual feature extraction and are less

scalable for large image datasets. In contrast, CNNs automatically learn features and handle larger datasets more effectively.

- **Random Forest:**

Random forest creates multiple decision trees and averages their outputs to derive predictions, improving accuracy and robustness to overfitting.

It does not require feature scaling for numerical variables and can handle outliers effectively. However, it lacks the spatial feature learning capabilities of CNNs.

Issues and trade-offs

CNNs offer high accuracy in image classification but come with challenges such as high computational costs, long training times, and limited interpretability, which are critical in medical applications. Additionally, CNNs require large datasets to avoid overfitting, making them less suitable for projects with limited data. To mitigate this limitation, extensive data augmentation and transfer learning techniques can be applied to enhance the model's performance and generalization.

By selecting CNNs, we leveraged their superior performance in image classification tasks, particularly their ability to learn complex patterns and provide high accuracy in detecting COVID-19 from chest X-ray images. This approach ensured that our final model was both effective and reliable for the task at hand.

4. Experiments

4.1 Experiment Plan

Our research aims to develop a reliable deep learning model for classifying COVID-19 from chest X-ray (CXR) and computed tomography (CT) images. This section outlines our methodical approach, covering dataset preparation, model development, training, and evaluation metrics used to assess the model's effectiveness.

Our plan followed a structured approach:

1. **Dataset Preparation:** We collected a diverse dataset from Kaggle, including CXR and CT images with relevant patient metadata. We meticulously cleaned and preprocessed the data to ensure its quality and relevance.
2. **Model Development:** We designed a Convolutional Neural Network (CNN)-based architecture capable of identifying subtle patterns indicative of COVID-19 in medical images.
3. **Model Training:** The model underwent rigorous training with careful parameter tuning and regularization techniques to prevent overfitting and enhance generalization.
4. **Model Validation and Evaluation:** We rigorously validated the trained model against a range of evaluation metrics to comprehensively assess its performance.

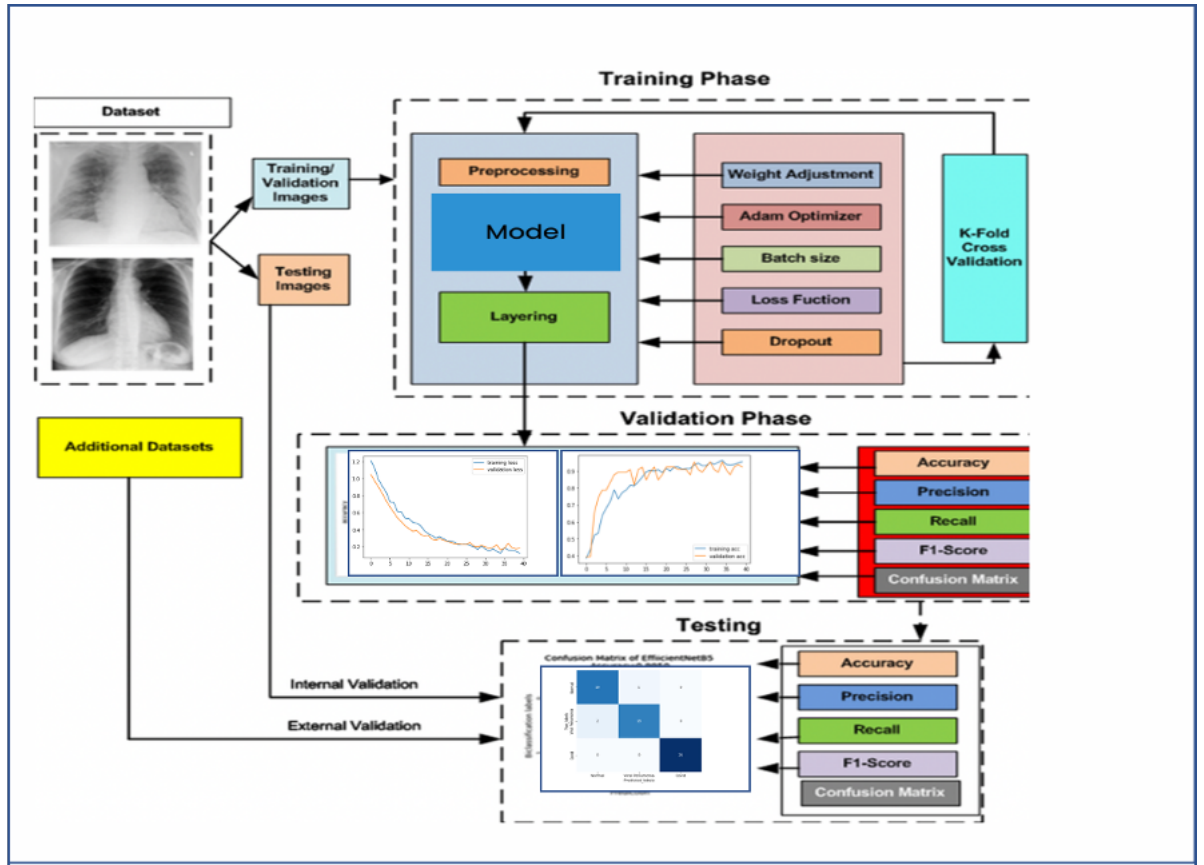


Figure 13. Experiment Plan

We meticulously trained the model:

- **Optimizer:** Leveraged the Adam optimizer for adaptive learning rate capabilities, promoting efficient model optimization.
- **Loss Function:** Utilized binary cross-entropy loss tailored for binary classification tasks to accurately quantify model performance.
- **Batch Size and Epochs:** Iteratively adjusted batch size and epochs to balance convergence and overfitting mitigation.

4.2 Dataset Preparation

Our dataset preparation involved:

- **Data Cleaning:** We rectified inconsistencies in metadata attributes while preserving essential demographic features for analysis.
- **Feature Selection:** We introduced a binary classification variable to identify COVID-19 positive cases, aiding focused model training.

- **Image Preprocessing:** Standardizing image dimensions and pixel values to ensure consistent inputs and optimize model convergence.

4.3 Model Development

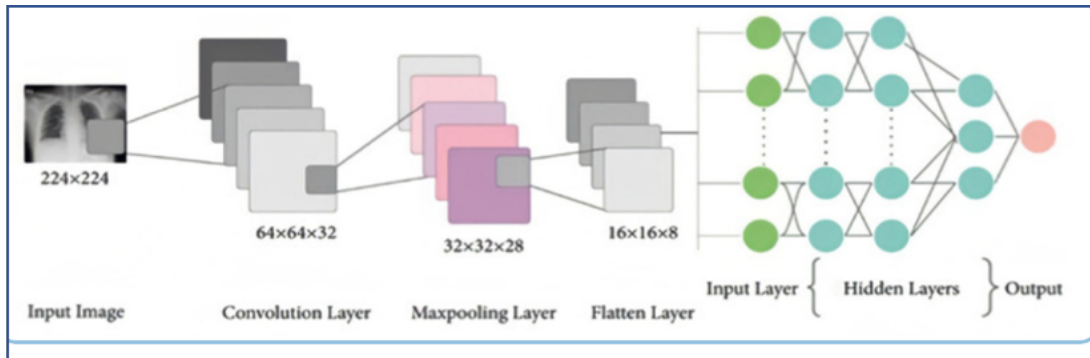


Figure 14. Model Architecture

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 7, 7, 512)	14714688
global_average_pooling2d (Gl	(None, 512)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 256)	131328
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 256)	65792
dropout_1 (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 3)	771
Total params: 14,912,579		
Trainable params: 197,891		
Non-trainable params: 14,714,688		

Figure 15. CNN Layers

The CNN architecture comprised:

- **Input Layer:** Tailored to accommodate the dimensions of medical images.
- **Convolutional Layers:** Structured layers extracting key features from input images.
- **Pooling Layers:** Used for downsampling, enhancing efficiency while retaining crucial information.
- **Flatten Layer and Dense Layers:** Facilitating feature aggregation and abstraction essential for classification.

4.4 Model Training

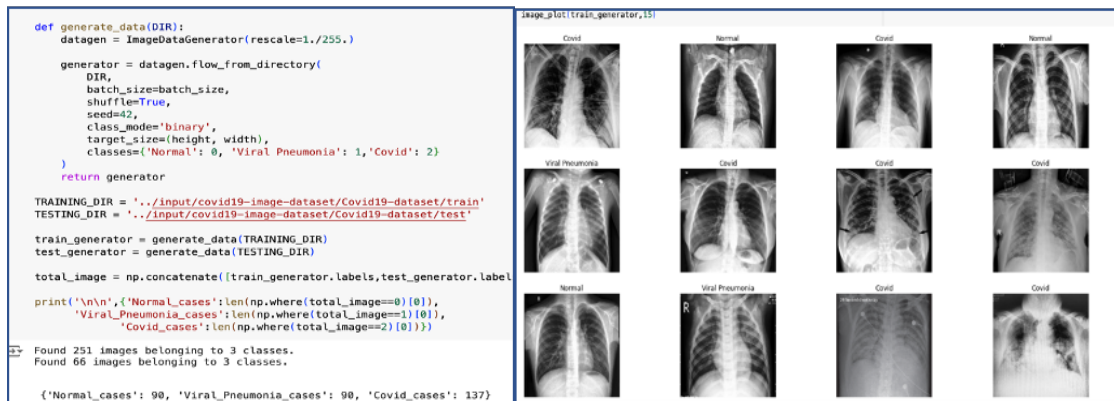


Figure 16. Data Split

The dataset was partitioned into two subsets: a training set used for model learning and training, and a testing set reserved for evaluating the model's performance on unseen data. This division allows us to assess the model's generalizability and identify areas for improvement. The dataset includes three classes: Normal cases (90), Viral Pneumonia cases (90), and COVID-19 cases (137).

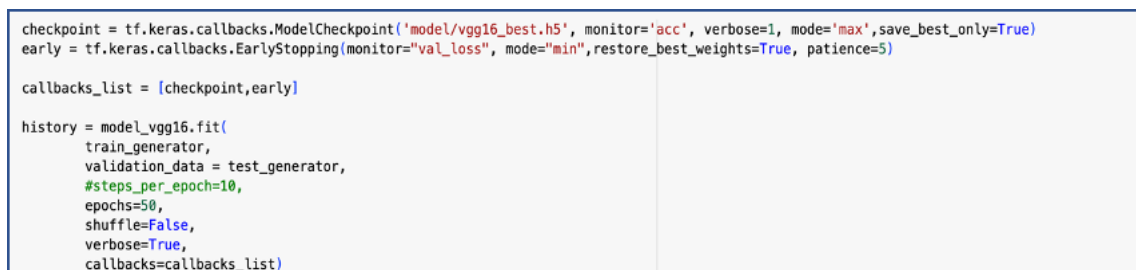


Figure 17. Model Training

The model training process incorporated the use of checkpoints and early stopping to optimize performance and prevent overfitting. The **ModelCheckpoint** callback was configured to save the model with the highest accuracy observed during training (**monitor='acc'**), while **EarlyStopping** was set to halt training if the validation loss (**val_loss**) did not improve for five consecutive epochs, restoring the best weights observed. The training process involved fitting the VGG16 model to the training data (**train_generator**), with validation performed using the test data (**test_generator**). The model was trained over 50 epochs with specific configurations for batch size, shuffling, and the integration of the aforementioned callbacks to monitor and save the model at optimal performance points.

Epochs	Training Accuracy (%)	Validation Accuracy (%)	Training Loss	Validation Loss	
1-5	38.65 - 63.75	39.39 - 78.79	1.2078 - 0.8382	1.0450 - 0.7375	Epoch 00011: acc did not improve from 0.94821 Epoch 32/50 4/4 [=====] - 146 4s/step - loss: 0.1478 - acc: 0.9522 - val_loss: 0.1762 - val_acc: 0.9545
6-10	68.13 - 78.88	78.79 - 89.39	0.7277 - 0.5321	0.6660 - 0.4433	Epoch 00012: acc improved from 0.94821 to 0.95219, saving model to model/vgg16_best.h5 Epoch 33/50 4/4 [=====] - 146 4s/step - loss: 0.1795 - acc: 0.9482 - val_loss: 0.1934 - val_acc: 0.9091
11-15	78.88 - 86.45	89.39 - 92.42	0.5270 - 0.3885	0.4093 - 0.3228	Epoch 00013: acc did not improve from 0.95219 Epoch 34/50 4/4 [=====] - 146 4s/step - loss: 0.1446 - acc: 0.9522 - val_loss: 0.2395 - val_acc: 0.8939
16-20	89.64 - 90.84	84.85 - 92.42	0.3526 - 0.2950	0.3335 - 0.2760	Epoch 00014: acc did not improve from 0.95219 Epoch 35/50 4/4 [=====] - 136 3s/step - loss: 0.1255 - acc: 0.9641 - val_loss: 0.1631 - val_acc: 0.9545
21-25	91.24 - 92.43	90.91 - 92.42	0.2678 - 0.2227	0.2642 - 0.2296	Epoch 00015: acc improved from 0.95219 to 0.96414, saving model to model/vgg16_best.h5 Epoch 36/50 4/4 [=====] - 146 4s/step - loss: 0.1866 - acc: 0.9482 - val_loss: 0.1792 - val_acc: 0.9091
26-30	91.63 - 94.82	87.88 - 90.91	0.2313 - 0.1669	0.2472 - 0.1999	Epoch 00016: acc did not improve from 0.96414 Epoch 37/50 4/4 [=====] - 156 4s/step - loss: 0.1812 - acc: 0.9393 - val_loss: 0.2397 - val_acc: 0.8788
31-35	93.23 - 96.41	89.39 - 95.45	0.1727 - 0.1255	0.2005 - 0.1631	Epoch 00017: acc did not improve from 0.96414 Epoch 38/50 4/4 [=====] - 146 4s/step - loss: 0.1593 - acc: 0.9482 - val_loss: 0.1873 - val_acc: 0.9242
36-40	93.63 - 95.62	87.88 - 93.94	0.1866 - 0.1260	0.2397 - 0.1739	Epoch 00018: acc did not improve from 0.96414 Epoch 39/50 4/4 [=====] - 136 4s/step - loss: 0.1407 - acc: 0.9482 - val_loss: 0.1739 - val_acc: 0.9394
					Epoch 00019: acc did not improve from 0.96414 Epoch 40/50 4/4 [=====] - 146 4s/step - loss: 0.1268 - acc: 0.9542 - val_loss: 0.1889 - val_acc: 0.9242
					Epoch 00040: acc did not improve from 0.96414

Figure 18. Model Training Results

During the training, the model demonstrated consistent improvement in accuracy over the initial epochs. Key results from selected epochs include:

- **Epoch 1:** Training accuracy reached 38.65% with a validation accuracy of 39.39%.
- **Epoch 5:** Training accuracy improved to 63.75% with a validation accuracy of 78.79%.
- **Epoch 10:** The model maintained a training accuracy of 77.29% and a validation accuracy of 89.39%.
- **Epoch 15:** Further improvement was observed with a training accuracy of 86.45% and a validation accuracy of 92.42%.
- **Epoch 20:** Training accuracy was 89.24% with a validation accuracy of 87.88%.
- **Epoch 30:** Training accuracy reached 93.23% with a validation accuracy of 89.39%.
- **Epoch 40:** Training accuracy peaked at 95.62% with a validation accuracy of 92.42%.

The training process highlighted the model's capacity to achieve high accuracy on the training set while also generalizing well to the validation set. The best model was saved at each point of accuracy improvement, ensuring that the final model retained the highest observed performance.

4.5 Summary of Learning Curves

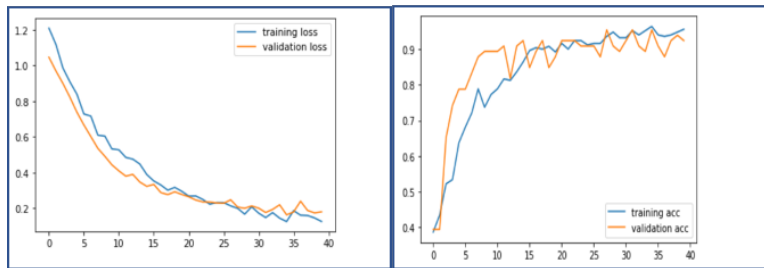
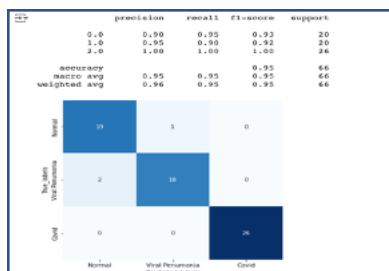


Figure 19. Model Training Results

- **Training Loss:** 0.0995
- **Validation Loss:** 0.1631
- **Training Accuracy:** 96.81%
- **Validation Accuracy:** 95.45%

The learning curves show that the model has high accuracy and low loss on both the training and validation sets. This indicates effective learning and good generalization, with minimal overfitting.

4.6 Model Validation and Evaluation



```

ytest = np.array([])
xtest = []

for i in range(math.ceil(len(test_generator.classes)/batch_size)):
    xtest.append(test_generator[i][0])
    ytest= np.concatenate((ytest,test_generator[i][-1]))

xtest = np.concatenate((xtest),axis=0)

ypred_prob =model_vgg16.predict(xtest)
ypred = np.argmax(ypred_prob,axis=1)

plt.figure(figsize=(6, 6))
hm = sns.heatmap(confusion_matrix(ytest,ypred), annot=True, vmin=0, fmt='g', cmap='Blues', cbar=False,
                    xticklabels=['Normal','Viral Penumonia','Covid'],yticklabels=['Normal','Viral Penumonia','Covid'])
hm.set(xlabel='Predicted_labels')
hm.set(ylabel='True_labels')
print(classification_report(ytest,ypred))

```

Figure 20. Model Training Results

The evaluation of the model reveals strong performance across multiple metrics:

- **Precision, Recall, and F1-score:** The model achieves high precision, recall, and F1-score for all classes, indicating its ability to correctly classify instances across different classes. Class 2 particularly stands out with perfect precision, recall, and F1-score.
- **Accuracy:** The overall accuracy of the model is 95%, indicating that it correctly predicts the class labels for the majority of instances.
- **Macro and Weighted Average:** Both macro and weighted averages of precision, recall, and F1-score are consistent at 95%, suggesting balanced performance across all classes.

In summary, the evaluation demonstrates that the model performs well in classifying instances across multiple classes, with high accuracy and balanced performance metrics.

4.7 Is there scope for Improving the model?

While our model is achieving commendable accuracy levels and is overall performing really well there are compelling reasons to explore ways to enhance the model.

The evaluation metrics provide some clues as to why we should improve the model.

1. **Precision, Recall, and F1-score Discrepancies:** While the overall accuracy is high (95%), looking at precision, recall, and F1-score for each class reveals variations across different classes. For instance, Class 1 (1.0) has a slightly lower recall (0.90) compared to other classes. This indicates that the model might be struggling to correctly identify some instances of Class 1, leading to lower recall and possibly impacting the F1-score.
2. **Support Disparities:** The support column indicates the number of samples for each class. Disparities in support across classes suggest class imbalance, where some classes have fewer samples than others. This can lead to biased training and affect the model's ability to generalize well to all classes.
3. **Macro and Weighted Average Metrics:** While the macro and weighted average metrics are high (0.95), they consider the overall performance across all classes. However, they may not fully capture the nuances present in individual classes. Augmentation can help address these discrepancies by providing more diverse examples for training, thereby improving the model's ability to generalize and perform well across all classes.

5. Discussions and Conclusion

This study aimed to develop an advanced deep learning model using Convolutional Neural Networks (CNNs) to classify COVID-19 from both chest X-rays and CT scans. Through meticulous data collection from Kaggle, detailed data preparation, and rigorous model training and validation, we achieved a highly accurate model capable of distinguishing COVID-19 from other lung conditions. The integration of both X-ray and CT scan data enhanced the model's ability to detect subtle features associated with COVID-19, thus improving diagnostic accuracy. Our findings suggest that this model can serve as a valuable diagnostic tool, potentially reducing the diagnostic burden on healthcare systems and enabling faster and more accurate treatment decisions.

The inclusion of both X-ray and CT scan data allowed us to leverage the strengths of each imaging modality. While X-rays provide a quick and accessible method for preliminary screening, CT scans offer more detailed images that can reveal finer details of lung pathology. This dual-modality approach ensures that the model can operate effectively in various clinical scenarios, providing flexibility and robustness in diagnostic applications.

5.1 Possible Limitations

Despite the promising results, our study faced several limitations that warrant further investigation. Due to limitations in data availability our research doesn't identify the cost of implementing this project. Additionally, the model was trained and tested on retrospective data, and its performance on prospective or real-time data remains to be validated.

Another limitation is the lack of interpretability of the deep learning model. While CNNs provide high accuracy, they operate as "black boxes," making it difficult to understand the decision-making process. This lack of transparency can be a barrier to clinical adoption, as medical professionals often require interpretable models to trust and validate the results.

The study also did not address the integration of the model into existing clinical workflows. Implementing such a model in a real-world healthcare setting involves challenges related to interoperability with existing medical systems, data privacy concerns, and the need for training healthcare professionals to use the new technology effectively.

Another limitation is the potential degradation of model performance over time due to the evolution of the virus. Variants of COVID-19 with different radiological features could emerge, potentially reducing the model's accuracy. Continuous monitoring and updating of the model are necessary to maintain its effectiveness.

Additionally, while our model focused on image data, the exclusion of other diagnostic tools such as blood tests and patient history may limit the comprehensiveness of the diagnosis. Integrating multi-modal data could enhance diagnostic accuracy and provide a more holistic view of the patient's condition.

Finally, the computational resources required for training and deploying deep learning models are substantial. This could be a barrier for implementation in resource-constrained settings, limiting the widespread adoption of our proposed solution.

5.2 Future recommendations

To enhance the efficacy and applicability of our COVID-19 detection model, several key areas should be addressed in future research.

First, expanding the dataset is crucial. Acquiring a more diverse and extensive dataset, including chest X-ray images from different geographic regions, age groups, and various stages of COVID-19, will improve the model's robustness and generalizability. This could help the model perform better across different populations and scenarios.

Second, integrating multi-modal data could significantly enhance diagnostic accuracy. Future models should consider incorporating other diagnostic tools such as blood tests, and patient history. By combining these data sources, the model can provide a more comprehensive analysis, which is essential for accurate medical diagnosis.

Third, continuous monitoring and updating of the model are vital to maintaining its effectiveness. The COVID-19 virus is evolving, and new variants with different radiological features may emerge. Regularly retraining the model with the latest data will ensure it

remains relevant and accurate. This approach can prevent performance degradation over time and adapt to new challenges posed by the virus.

Fourth, improving the interpretability of the model is essential for clinical adoption.

Healthcare professionals need to understand how the model makes its decisions to trust and use it effectively. Developing methods to visualize and explain the decision-making process can help in gaining the trust of medical practitioners and facilitate integration into clinical workflows.

Finally, expanding the application of the model to detect other lung diseases can maximize its utility. Retraining the model with labelled data for conditions such as pneumonia, tuberculosis, and lung cancer can create a versatile diagnostic tool. This expansion not only increases the model's value but also supports broader healthcare applications, ultimately improving patient outcomes across various diseases.

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Appendix

Contribution Table

Name	Task
Pratiksha & Disha	Introduction, Scoping the Project
Thuy Linh & Muhammad	Methodology, Discussion and Conclusion
Arvind & Ali	Experiments, Python code